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Simultaneous Estimation of Engine Deck and Drag Polar Based on Specific Excess Power Measurement

Fernando A. Domingos* and Flávio Luiz Cardoso Ribeiro†
ITA - Instituto Tecnológico de Aeronáutica, São José dos Campos, São Paulo, 12228-900, Brazil

Bruno Giordano de Oliveira Silva‡
IPEV - Instituto de Pesquisas e Ensaios em Voo, São José dos Campos, São Paulo, 12228-904, Brazil

Most performance data reduction methods rely on the availability of a flight thrust deck. Nevertheless, for many reasons, such as industrial intellectual property, this information is not always available for the aircraft end user, or has prohibitive costs. For that reason, the ability to estimate an engine flight thrust deck based on flight data may be of special interest for flight test organizations or flight test schools. Thus, the objective of this work is to demonstrate a flight test method that enables the estimation of a reliable engine deck, using limited flight test instrumentation. The proposed method uses the specific excess power to estimate the thrust and drag balance. By combining level flight accelerations at constant altitudes and climb/descents at constant Mach numbers, using different thrust settings, it was possible to estimate the engine thrust for a combination of engine rotation speeds, altitudes, and airspeed. Using the method, it was also possible to estimate the drag polar at different Mach numbers.

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I. Nomenclature

D	=	Drag force, N
W	=	Weight, N
L	=	Lift force, N
n_z	=	Normal load factor
T_n	=	Net thrust, N
F_T	=	Tangential net thrust component, N
N_1	=	Fan speed, %rpm
N_2	=	Compressor speed, %rpm
x_a, y_a, z_a	=	Aerodynamic axes, m
x_e, y_e, z_e	=	Earth referenced axes, m
$F_{x,\,\mathrm{aed}}$	=	X component of the resultant pressure force acting on the vehicle, N
$F_{z,\,\mathrm{aed}}$	=	Z component of the resultant pressure force acting on the vehicle, N
V	=	True airspeed, m/s
f, g	=	Generic functions
h	=	Height, m
M_a	=	Mach number
S	=	Aerodynamic reference area, m^2
$\dot{H}$	=	Specific excess power, m/s
m_{af}	=	Aircraft mass, kg
A	=	Propulsion reference area, m^2
$\dot{w}$	=	Fuel mass flow, kg/s
γ	=	Flight path angle, deg

*Undergraduate student, Instituto Tecnológico de Aeronáutica, Praça Mal. Eduardo Gomes 50, São José dos Campos, domingosfad@fab.mil.br.
†Assistant Professor, Instituto Tecnológico de Aeronáutica, Praça Mal. Eduardo Gomes 50, São José dos Campos, flaviocr@ita.br.
‡Flight Test Engineer, Instituto de Pesquisas e Ensaios em Voo, Praça Mal. Eduardo Gomes, 50, São José dos Campos, brunogiordanobgos@fab.mil.br.

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II. Introduction

Most methods for performance data reduction rely on the availability of an extensive set of flight thrust data. However, for various reasons, such as intellectual property protection, this information is not always available to the aircraft operator or has prohibitive acquisition costs. This absence of design models hinders the study and advanced research on aircraft, as well as limits its ability to serve as an appropriate platform for studies. However, to facilitate future trials and studies with the aircraft, it is of utmost importance to perform modeling and parametrization of all the aircraft's flight behavior. Through this modeling, it will be possible to obtain essential information for performance analysis, aerodynamic characteristics, and other aspects relevant for flight safety analyses.

One of the first steps in analyzing a new aircraft is to have knowledge of its performance characteristics. These main characteristics are the engine thrust deck and drag polar. The thrust deck is a set of tables or a computer program that presents the thrust values corresponding to the aircraft current flight conditions. Knowing the thrust deck, it is possible to understand and optimize the aircraft performance in different flight conditions. This information will be especially useful for analyzing and predicting the aircraft behavior following modifications to its configuration, such as the addition of external devices or structural changes. Direct measurement methods of thrust during flight require significant modifications to the aircraft, and establishing reliable calibration and measurement procedures is also a challenging task. While the known thrust estimation methods require specific instrumentation not normally present in serial aircraft, making it difficult for performance analysis, especially in commercial aircraft.

Turbofan engines are commonly used in commercial or executive aircraft. Estimating in-flight thrust of turbofan engines is a complex task, as the values depend on various variables [1]. Some variables are well known and determined, such as the aircraft's altitude, the Mach number at the inlet, and the position of the thrust levers. However, other variables are often unknown, have inaccurate estimates, or depend on invasive and expensive instrumentation, such as the temperature differential in the turbine and the engine's level of degradation.

One of the earliest methods for determining in-flight thrust was the Mass Consumption Acceleration (MCA)[2], which is based on obtaining a drag polar curve and from it estimating in-flight thrust. The analysis of this method proposes obtaining an efficiency function associated with the conversion of fuel energy into total mechanical energy. By estimating parameters in both the efficiency and the drag polar, it becomes possible to estimate in-flight thrust.

The next effort was the Excess Force and Mass Acceleration Method (EFMA) [3], derived from the MCA and developed in 2001 by the same author of the previous study. This new method uses a sophisticated database, employing data on speed, alpha and beta, inertia, and positioning. The physical principle of the EFMA is to calculate the excess force using the mass and acceleration measurements of the aircraft. The estimated thrust value is obtained by analyzing the aircraft's acceleration behavior in different power configurations.

In both studies, the aircraft are equipped with specialized instrumentation systems, installed specifically for flight test and research purposes. The proposal to use commercial aircraft for flight studies and tests underscores the need to develop a method capable of assessing an aircraft's performance characteristics using the standard instrumentation found in modern commercial aircraft. The ability to conduct flight tests without the need for specialized instrumentation significantly reduces financial and time expenses.

A significant recent advancement in the field was made by Deiler's work [4] on engine thrust identification. The study decorrelated drag and thrust on a large database of operational flights with the Airbus A320neo aircraft. This was achieved using a similar approach centered on an energy perspective of the aircraft, particularly focusing on the aerodynamic variance due to changes in flap/slat configurations. Despite the limited information about the actual engine thrust available in the data, the models employed were adept at accurately predicting the engine thrust characteristics within the required flight envelope.

In the present work, the simultaneous estimation of drag polar and engine thrust parameters is achieved by means of quasi-steady maneuvers, involving climbs, descents and level accelerations and deccelerations performed at fixed thrust lever positions. The flight data is obtained from onboard measurements read post-flight from the Flight Data Recorder (FDR) system. Polynomial functions are used to represent drag polar and engine thrust for a set of reference Mach numbers, and their coefficients are estimated using a Least-Squares optimization on the measured versus modeled specific excess power.

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III. Aero-Propulsive and Energy Model

A. Aircraft Equations of Motion In Longitudinal Flight

Assuming that the aircraft is in a climb regime with negligible variation in climb speed. As an approximation, we consider small climb angles. The diagram of forces acting on the aircraft in this flight regime is shown in Figure 1, where T_n is the total net thrust, V is the true airspeed, α is the angle of attack, and γ is the flight path angle, L and D are the aerodynamic lift and drag forces, respectively.

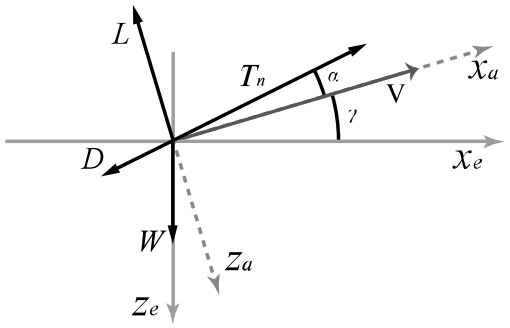


Fig. 1 Force diagram in longitudinal flight.

Decomposing the respective forces into the aerodynamic coordinate system according to the scheme of Figure 1, we obtain the following equations:

$$\sum F_{x,\,\text{aed}} = T_n \cos(\alpha) - D - W\sin(\gamma) = \frac{W}{g} \cdot \dot{V} \tag{1}$$

$$\sum F_{z,\,\text{aed}} = W\cos(\gamma) - L - T_n\sin(\alpha) = \frac{W}{g} \cdot V \cdot \dot{\gamma} \tag{2}$$

B. Aircraft Equations of Energy

For the same system, we can analyze mechanical energy, in which the change in mechanical energy in the rigid body is given by the work done by the resultant force acting externally on the aircraft. Considering that the force acting on the aircraft is the excess propulsion force, given by the relation of aerodynamic resultant force in x, by equation 1. The translational mechanical energy of a rigid body is given by:

$$E = m_{\text{ac}}\frac{V^2}{2}$$

Deriving the energy over time, we get the power:

$$\frac{d\,E}{d\,t} = m_{\text{ac}} V \frac{d\,V}{d\,t} \tag{3}$$

The work done by an external force is the integral of the force by the displacement, arranging terms in terms of external force we obtain:

$$\sum F_{\text{ext}} = \frac{d\,E}{d\,x} = \frac{d\,E}{d\,t}\,\frac{d\,t}{d\,x} \longrightarrow \frac{d\,E}{d\,t} = \left(\sum F_{\text{ext}}\right)\frac{d\,x}{d\,t}$$

Substituting the power relation 3 and the statics relation 1:

$$\left(\sum F_{\text{ext}}\right)\frac{d\,x}{d\,t} = m_{\text{ac}}\; g\left(\frac{V}{g}\frac{d\,V}{d\,t}\right)$$

$$m_{\text{ac}}\; g\left(\frac{V}{g}\frac{d\,V}{d\,t}\right) = \left(T_n\cos(\alpha) - D - W\sin(\gamma)\right)\frac{d\,x}{d\,t}$$

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In the equation, the term of the aircraft's forward motion rate over time represents the true airspeed of the aircraft, so:

$$m_{\text{ac}}\ g\left(\frac{V}{g}\frac{d\ V}{d\ t}\right) + W \cdot \sin(\gamma) \cdot V = (T_n \cos(\alpha) - D) \cdot V$$

$$m_{\text{ac}}\ g\left(\frac{V}{g}\frac{d\ V}{d\ t} + \frac{d\ h}{d\ t}\right) = (T_n \cos(\alpha) - D) \cdot V \tag{4}$$

In Equation 4, the term on the left represents the change in specific energy of the body with respect to time, which represents a power. Isolating the weight, we obtain the expression for specific power:

$$\dot{H} = \frac{V}{g}\frac{d\ V}{d\ t} + \frac{d\ h}{d\ t} \tag{5}$$

As another simplification, we can analyze thrust as a tangential force component under specific flight conditions: $F_T = T_n \cos(\alpha)$

Substituting in Equation 4 the expression 5, and also dividing the terms by the weight W, we obtain:

$$\frac{\dot{H}}{V} = \left(\frac{F_T - D}{W}\right) \tag{6}$$

Based on Equation 5, by knowing the temporal values of speed, altitude, and mass, it becomes feasible to calculate the energetic behavior of the aircraft at each moment of the flight. This information is crucial for the proposal of a system identification model, which employs the specific power relation to simulate the energetic behavior of the aircraft in longitudinal flight.

As detailed in Sections III.C.2 and III.C.1, the thrust and drag models are integrated into the analysis. Using Equation 6, in conjunction with these models, it is possible to adjust the corresponding parameters, ensuring that the modeled energetic behavior closely approximates the reality observed in flight tests. The system identification method that will use this energy relation will be detailed in Section V.

C. Aircraft Engine Aero-Propulsive Model

A primary requirement for any engine thrust model is that it should be smooth and without significant local changes in curvature. Analyses demonstrated by parametric models in III.C.1 show no discontinuous or different behavior of thrust with fan rotation, speed, altitude, or expected pressure [4]. Furthermore, engineering knowledge about jet engines and how they produce thrust directly imposes the need for smooth thrust curves: a jet engine, within its normal operational limits, will not produce an abrupt change in thrust due to a change in flight condition because the thermodynamic cycle in the engine is operating continuously [5]. Thus, these models do not model disturbances or transition conditions, such as engine stall, abrupt changes in throttle, or any malfunctions.

1. Engine Thrust Model

Following the numerical thrust models previously discussed, a hybrid model was proposed between a thrust chart with lookup table and data interpolation for a thrust parameter chart and obtained for the 4 Mach numbers from which experimental flight data were obtained.

The chosen parametric model was the polynomial without explicit Mach parameters, as the Mach dependency will be determined by the variation in the parameters.

$$\frac{T_n}{\delta} = f\left(\frac{N_1}{\sqrt{\theta}}\right)$$

$$\frac{T_{n,E}}{\delta} = T_{E,0} \cdot \left[P_0 + P_{N1}\left(\frac{N_1}{\sqrt{\theta}} - 1\right) + P_{N2}\left(\frac{N_1}{\sqrt{\theta}} - 1\right)^2\right] \tag{7}$$

Which were tested at 4 Mach numbers:

Thrust Parameters

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$$T_P = \begin{bmatrix} P_{0,1} & P_{0,2} & P_{0,3} & P_{0,4} \\ P_{\mathrm{N1},1} & P_{\mathrm{N1},2} & P_{\mathrm{N1},3} & P_{\mathrm{N1},4} \\ P_{\mathrm{N2},1} & P_{\mathrm{N2},2} & P_{\mathrm{N2},3} & P_{\mathrm{N2},4} \end{bmatrix}$$

Therefore, the function used for optimization, in which the behavior of Mach number variation will be embedded in T_P, is:

$$\frac{T_n}{\delta} = f\left(T_p, \frac{N_1}{\sqrt{\theta}}\right) \tag{8}$$

2. Aircraft Drag Model

Adopting the simplest aerodynamic drag model, the proposal focused on using only parasitic and induced drag parameters, which were deemed most significant in assessing the aircraft's performance. However, it was also considered that introducing more complex parameters into the drag component might lead to an undesirable coupling between the thrust and drag models. Such coupling could potentially impede the modeling capabilities. This consideration underlines the importance of maintaining simplicity in the models, which aids in effectively identifying and decoupling the coplanar forces of drag and thrust.

$$\frac{D}{\delta} = g\left(M_a, \frac{W}{\delta}\right)$$

$$\frac{D}{\delta} = \mathrm{Cd}_0 \frac{P_0\, S_{\mathrm{ref}}\, \gamma}{2} {M_a}^2 + k \frac{2}{P_0\, S_{\mathrm{ref}}\, \gamma} \frac{W^2}{\delta^2} \frac{1}{{M_a}^2} \tag{9}$$

A similar proposition to the thrust model was made for the drag model, in which a table of parameters related to the aircraft's drag coefficients for each of the Mach numbers tested was created.

Drag Parameters

$$D_C = \begin{bmatrix} k_1 & k_2 & k_3 & k_4 \\ \mathrm{Cd}_{01} & \mathrm{Cd}_{02} & \mathrm{Cd}_{03} & \mathrm{Cd}_{04} \end{bmatrix}$$

Therefore, the function used for optimization, in which the behavior of Mach number variation will be embedded in D_C, is:

$$\frac{D}{\delta} = g\left(D_C, M_a, \frac{W}{\delta},\right) \tag{10}$$

IV. Flight Database

A. Test Vehicle

Characteristics of the U-100 Phenom aircraft.[6]

- Wingspan: 12.3 m
- Wing Area: 18.76 m²
- Empty Weight: 3,275 kg
- MTOW: 4,800 kg
- Fuel Capacity: 1,272 kg
- Powerplant: 2 x 7.7 kN
- Maximum Speed: Mach 0.7
- High Speed Cruise: 406 kt
- Minimum Control Speed: 100 kt
- Range: 1,178 NM



Fig. 2 Embraer Phenom 100, U-100

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B. Aircraft Internal Instrumentation available

The Phenom 100 aircraft is equipped with a state-of-the-art electronic system, incorporating functionalities that meet most standard operational requirements. However, for flight test purposes, this equipment is not entirely suitable. The data acquisition frequency, limited to just 1 Hz, is insufficient to capture all the nuances of the aircraft's flight behavior. Moreover, the accuracy of the data, the variety of monitored variables, and the data acquisition frequency may not align with the specific demands of a research project.

Table 1 presents a list of data provided by the internal data acquisition system of the Phenom 100, with each variable being detailed by its unit, associated symbol in the project, and corresponding precision. The listed variables are obtained through the Flight Data Recorder(FDR).

Table 1 U-100 FDR's Data

Variable	Unit	Symbol	Precision
time	[s]	t	10^0
Vel_{GPS}	[m/s]	V_e	0.1
Z_{GPS}	[m]	Z_e	0.3
Vc	[KCAS]	V_{kas}	0.05
Zp	[ft]	h	1
Mach	[adm]	M_a	0.001
N_Z	[m/s^2]	N_z	0.025
theta	[deg]	θ	0.005
phi	[deg]	ϕ	0.005
TAT	[C]	T_t	0.25
Pa	[mb]	P_a	$3 \cdot 10^{-2}$
E1 ITT	[C]	ΔT_1	0.5
E2 ITT	[C]	ΔT_2	0.5
E1 FF	[kg/h]	$\dot{w}_1$	3
E2 FF	[kg/h]	$\dot{w}_2$	3
E1 N1	[%]	$N_{1,1}$	0.1
E2 N1	[%]	$N_{1,2}$	0.1
E1 TLA	[%]	$\delta_{t,1}$	0.1
E2 TLA	[%]	$\delta_{t,2}$	0.1
E1 N2	[%]	$N_{2,1}$	0.1
E2 N2	[%]	$N_{2,2}$	0.1

C. Flight test maneuvers

1. Level Acceleration

Level Acceleration Maneuver: This maneuver primarily involves converting the aircraft's engine power into kinetic energy, leading to an increase in airspeed while maintaining a constant altitude. It's a direct assessment of the engine's efficiency in transforming fuel energy into propulsive force and gauging the aerodynamic efficiency of the aircraft at varying speeds. The focus is on how the aircraft balances energy input (from the engine) and energy loss (due to drag and other resistive forces).

2. Constant Mach Climbs and Descents

Constant Mach Climbs and Descents: In these maneuvers, the aircraft navigates a trade-off between kinetic and potential energy. During a climb, the engine's power is used to overcome gravity, converting kinetic energy into potential energy, thereby gaining altitude without altering airspeed. In a descent, potential energy is converted back into kinetic energy or lost through aerodynamic drag. These maneuvers are crucial for understanding the aircraft's efficiency in maintaining speed during altitude changes and its capability in managing energy transitions under different atmospheric conditions.

D. Flight Test Data

To assess the performance characteristics of the aircraft, a series of flight tests were carried out featuring a variety of maneuvers, including acceleration, vertical movements, and level turns. These tests were strategically conducted across different Mach number regimes, spanning a range of altitudes. Specifically, the trials targeted calibrated speeds corresponding to Mach numbers of 0.3, 0.42, 0.54, and 0.65.

It is important to note that test point 14, which was planned at an altitude of 40,000 feet, could not be stabilized and was consequently excluded from the analysis. This decision was driven by constraints on the total flight time available for the tests. In addition, to maximize the efficiency of the testing process within the time limits, the stabilization period for the maneuvers was reduced. Initially set at about 60 seconds for the lower altitude points, it was adjusted to 40 seconds for the higher altitude tests, starting from test point 11. This adjustment ensured that a wider range of test points could be covered effectively, while still providing adequate stabilization time for each maneuver.

The chart presented in Figure 3 illustrates the flight diagram with the proposed test points, incorporating the collected flight data.

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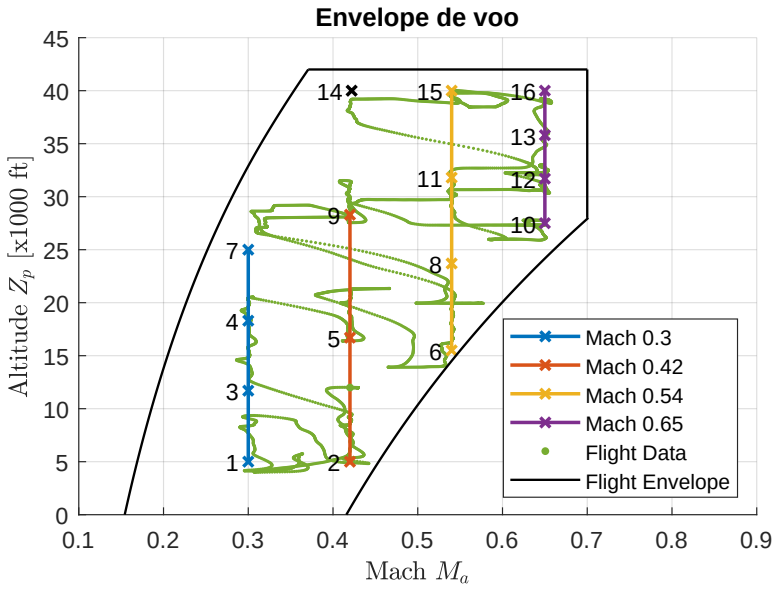


Fig. 3 Flight envelope with test points.

E. Data Partitioning

As an initial proposal, all flight points following the quality criteria for each type of maneuver were selected.

Climbs or descents maneuvers:

- Stabilized values of N_1.
- Mach speed close to the test points and stabilized.

Level accelerations and turns:

- Stabilized values of N_1.
- Constant throttle values.

The selection criteria for the flight data in this study were specifically designed to align with the capabilities of our engine model. Given that our model primarily represents the static behavior of the engine and is not equipped to accurately capture its dynamic responses, particularly changes in fan rotation speed, we placed a strong emphasis on choosing flight points with stabilized N_1 values. This approach ensures the data aligns closely with the static modeling framework, thereby enhancing the accuracy and reliability of our analysis. In essence, by focusing on data points where N_1 is stable, we can confidently apply our model to predict engine behavior within its static operational limits, despite its inability to interpret dynamic engine characteristics.

Following this decision-making scale, Figure 4 illustrates the progression of the flight, distinguishing between the data used for optimization, highlighted in orange, and the disregarded data, shown in blue.

V. System Identification and Parameter Estimation Algorithm

The system identification process begins with the selection of flight segments that contain useful data. Once selected, a low-pass filtering procedure is applied to the temporal data to mitigate the effect of high-frequency noise. Subsequently, weight estimates are calculated based on fuel consumption and fuel flow rate. The analysis proceeds with the execution of the derivatives of speed and altitude variation, in order to obtain the specific energy for each segment. The value of $\dot{H}$ is then calculated for each flight point, providing a vector of values $\dot{H}_{dados}$ representing the total energetic behavior of the system over time.

Sequentially, the data collected by the acquisition system are transformed into a suitable input vector for the functional relationship used for identification. The input vector becomes $X_{dados} = [N_1/\sqrt{\theta}, W, N_z, M_a, \delta_e]$.

$$\min_P \|\dot{H}(P, X_{dados}) - \dot{H}_{dados}\|_2^2 = \min_P \sum_i \left(\dot{H}(P, X_{dados})_i - (\dot{H}_{dados})_i\right)^2, \tag{11}$$

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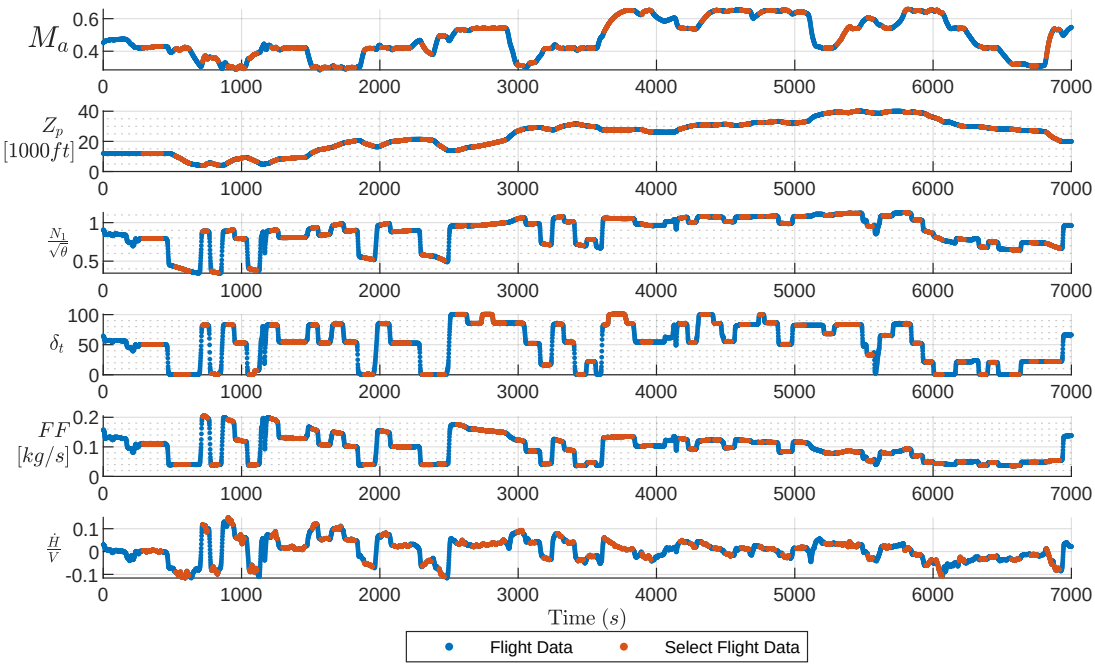


Fig. 4 Aircraft variables throughout the flight.

The identification process involves finding the model parameters that reduce the error between the value of $\dot{H}$ obtained from the data and the estimated by the model, providing a performance metric for the representation of the system, as represented in Equation 11.

The Flowchart in Figure 5 describes the step-by-step process of the system identification code.

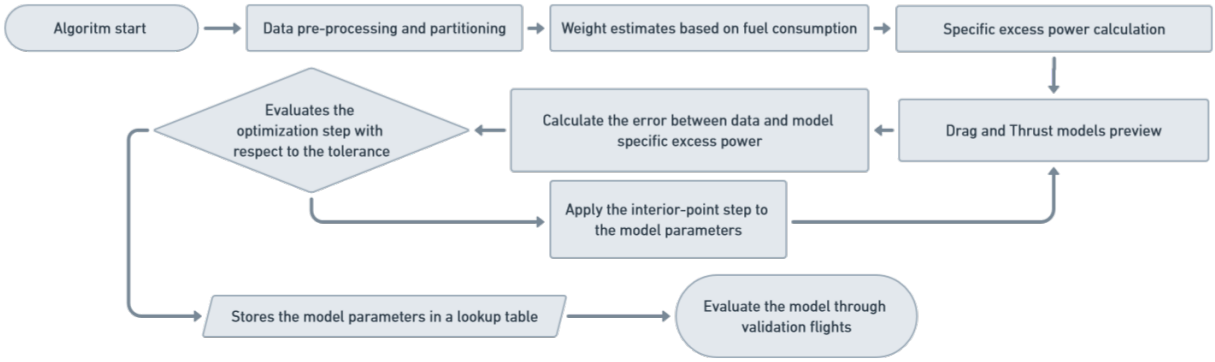


Fig. 5 Flowchart of the parameter estimation code

An interior point optimization step is then employed to refine the parameters that best represent the propulsive and aerodynamic systems for the analyzed flight. Ensuring that the parameters used comply with the imposed optimization rules, such as boundaries and constraints on the parameters. The analysis is concluded with the evaluation of the calculated error in comparison with other flight data, providing additional validation of the proposed model and the optimized parameters.

VI. Parameter Estimation Results

A. Specific Power Compatibility

Conducting system identification with the selected data yielded the following model for estimating specific power, depicted in Figure 6.

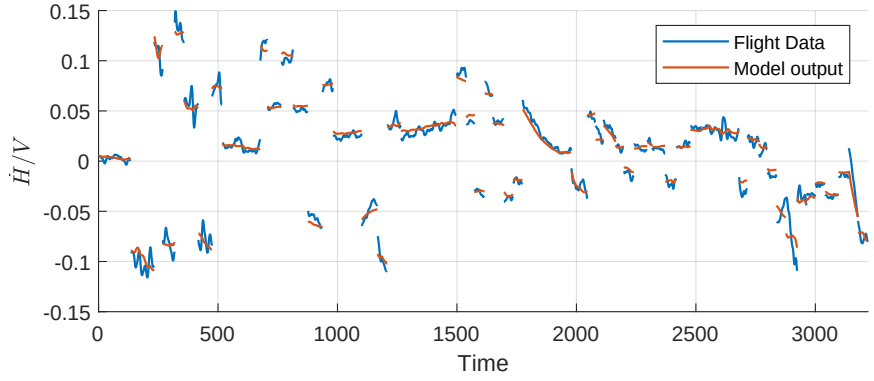


Fig. 6 Specific power over time.

The compatibility between real flight data and the model was deemed satisfactory, with the model achieving a determination coefficient of 96.7%. With this determination coefficient, it is evident that the model can effectively replicate a significant part of the aircraft's dynamic behavior during flight. However, notable deviations, particularly in high-frequency responses and transient conditions observed in flight data, were noted.

The deviations in high-frequency responses can be attributed to the simplified nature of the model, which, despite having a low-pass filter to attenuate high-frequency fluctuations, still provides a statistical approximation to the aircraft's behavior. This statistical approach is not sufficient to capture the nuances of flight dynamics. Similarly, during transient conditions, the flight dynamics become more complex, with various aerodynamic parameters and flight controls interacting non-linearly. The model is not capable of fully capturing these non-linear interactions, resulting in deviations in predictions.

Additionally, the interaction of the pilot with the aircraft is an aspect that can be challenging to model accurately. During flight, the pilot makes a series of small corrections to keep the aircraft stabilized under desired conditions. These continuous corrections, though minor, can significantly impact the aircraft's energy reactions, altering flight dynamics and, consequently, the model's accuracy in replicating the aircraft's behavior.

B. Decoupling of Thrust and Drag

The employed methodology allowed for discrimination between net thrust and drag. The values of the thrust and drag parameters found are provided in Table 2:

Table 2 Values of the parameters found.

Param.	$M.3$	$M.42$	$M.54$	$M.64$
P_0	0.9867	0.9674	0.9674	0.9679
$P_{N1,1}$	2.9117	2.8041	2.8922	3.4093
$P_{N1,2}$	2.1366	2.1525	2.3441	1.0367
k	0.0439	0.0439	0.0439	0.0439
Cd_0	0.0218	0.0260	0.0260	0.0274

The analysis of the table reveals significant aspects about the propulsive model and the aircraft's drag polar. The parameters P_0 and Cd_0, representing the initial propulsive efficiency and parasitic drag coefficient, respectively, show very close values. This proximity is attributed to the optimization constraints imposed on the model.

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The observed closeness suggests that the model is operating under significant constraints, indicating the need for data collection at various speeds to ensure that optimization reaches physically plausible values. Gathering data across a broader spectrum of speeds is essential to validate the model's efficacy and ensure its applicability in varied operational scenarios.

The value of the coefficient k, kept unchanged in all configurations, corresponds to the estimate of Oswald's efficiency factor, a crucial parameter in aerodynamic analysis. The model achieved an Oswald efficiency value of $e = 0.87$, which is in line with the expected average value for similar-sized aircraft equipped with wings of the same aspect ratio. This congruence suggests adequate calibration of the model in relation to the predominant aerodynamic characteristics of similar configuration aircraft.

Regarding the values of $P_{N1,1}$, a progressive increase is observed with the increment of the Mach number. This trend indicates a direct relationship between propulsive efficiency at high speeds and the Mach number, highlighting the influence of aerodynamic speed on propulsive performance. In contrast, the values of $P_{N1,2}$ demonstrate a general increasing trend, except for the term corresponding to the maximum Mach. This exception can be interpreted as an indication of variations in propulsive efficiency under extreme speed conditions, suggesting a differentiated behavior of the propulsive model in this operational regime.

In summary, the analysis of the parameters presented in the table provides valuable insights into the behavior of the propulsive model and the drag polar under different operational conditions, highlighting the importance of an integrated approach to performance optimization in aircraft.

Figure 7 depicts the expected values for the aircraft's thrust and drag at the test points of the model.

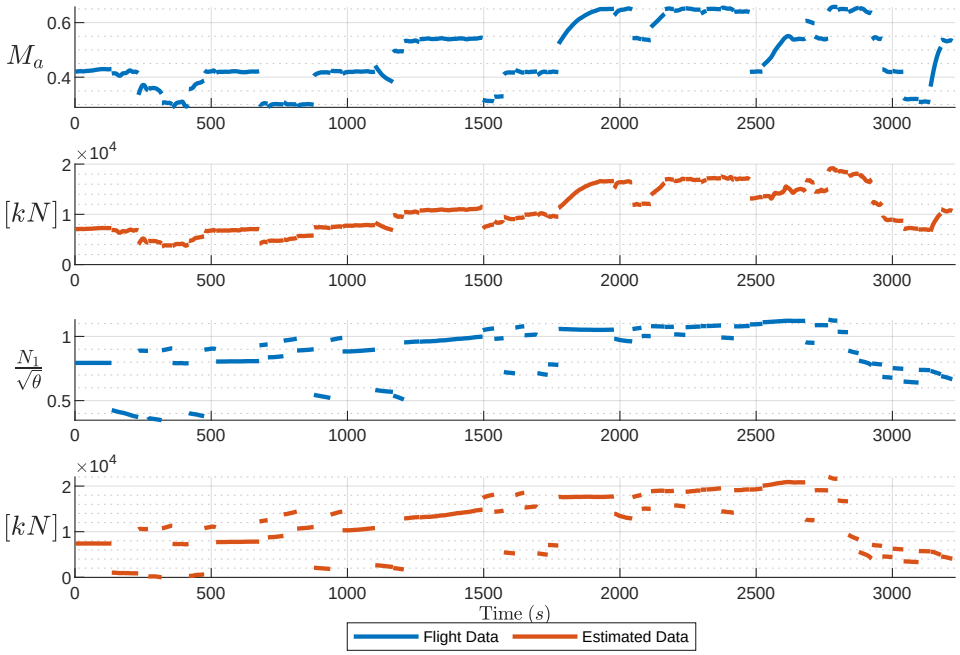


Fig. 7 Estimated Thrust and Drag of the aircraft for test data.

In this figure, it is evident that the thrust demonstrates a significant correlation with changes in the fan rotation variable. In parallel, drag exhibits a positive correlation with Mach number values. These correlations indicate that the model can effectively describe the aircraft's dynamic behavior.

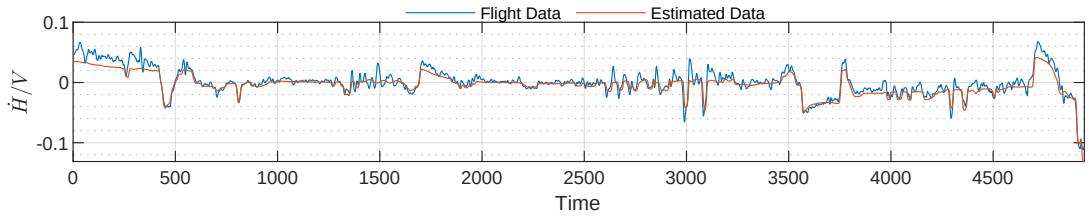
VII. Validation Tests

As the final step in the process, models trained only with data from the test flight were compared with data obtained from two other conventional flights of the aircraft.

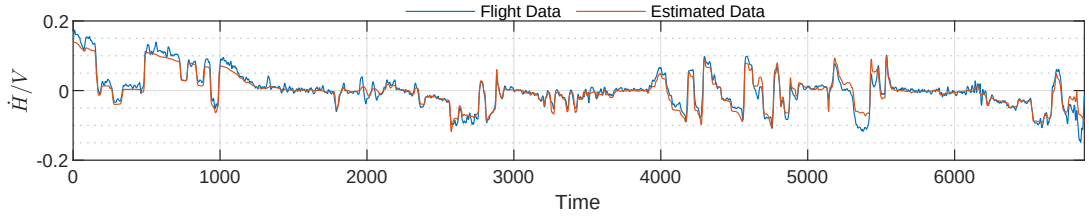
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A. Specific Power

System identification performed with the validation flights yielded the following specific power charts, depicted in Figure 8.



(a) Specific power for flight 1, $R_{E1} = 88.1\%$



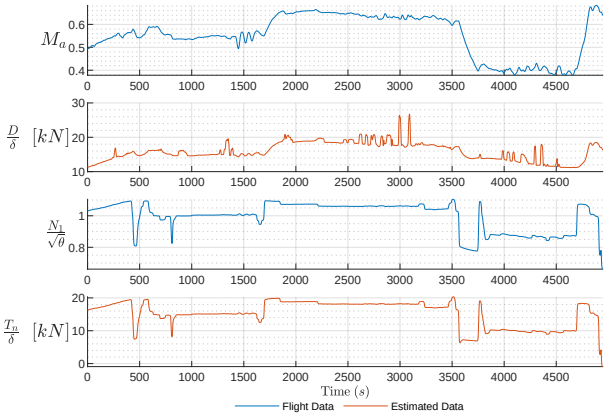
(b) Specific power for flight 2, $R_{E1} = 92.4\%$

Fig. 8 Estimation of specific power for the validation flights

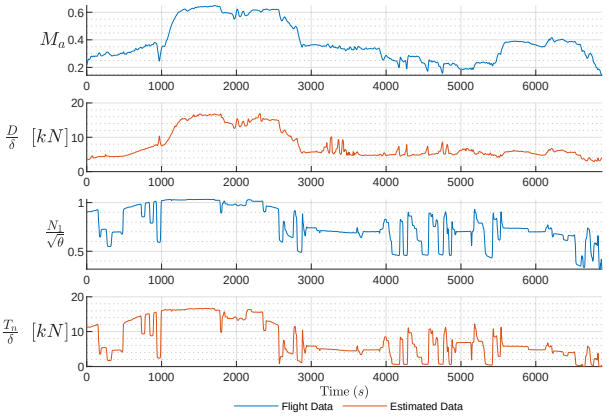
The comparative analysis of flight data with the model resulted in a determination coefficient of 88.1% for flight 1 and 92.4% for flight 2. Notable deviations were observed in both flights at certain points. These deviations could be attributed to a variety of factors not contemplated in the model, including, but not limited to, activation of the engine bleed system or manipulation of the flaps. As there is no information about the execution of the flights in question, it is not possible to assume origins regarding the causes of the observed deviations.

B. Drag and Thrust

Based on the thrust and drag models, the following traction and drag charts for the two validation flights were obtained:



(a) Thrust and Drag for flight 1



(b) Thrust and Drag for flight 2

Fig. 9 Estimation of Thrust and Drag for the validation flights

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VIII. Conclusions and Future Work

In this study, the Least Squares Method, using an interior-point optimization algorithm, was applied to model the performance of a Phenom 100 aircraft. The model's validation was evidenced by a consistent success rate in predicting specific power, exceeding 90%. This high success rate is particularly notable given the limited set of flight data used in calibrating the model, underscoring its effectiveness and the accuracy of the predictions provided.

In summary, the adopted methodology proved to be a viable and cost-effective approach for aircraft performance analysis, especially for those aircraft that do not accommodate modifications for the installation of dedicated measurement systems, as is the case with the equipment used by the Flight Test and Research Institute (IPEV). Therefore, the work presented provides a valuable contribution to the field of aerospace engineering, offering a reliable methodology for aircraft evaluation that dispenses with the need for structural changes or the addition of invasive measurement instruments.

For the future development of this project, one of the main goals is to collect a larger volume of test data, aiming to refine the process of identifying the components of the engine and the aerodynamic dynamics. Expanding the database will not only refine the current predictions but also extend the model's prediction envelope, covering a broader range of flight conditions and operational scenarios, thereby enhancing confidence in the aircraft's safety for initiating flight tests [7].

Particular focus will be given to the integration of additional parameters that may influence aircraft performance, such as engine temperature and variations in standard atmospheric temperature. The inclusion of these variables has the potential to correct minor deviations observed in the data, which can be attributed to differences in the environmental conditions under which the test flights are conducted.

Considering the twin-engine nature of the aircraft under study, it is intended to conduct tests that explore the condition of differential thrust. This method of differential analysis can more accurately decorrelate thrust and drag data, providing a more detailed understanding of the forces acting on the aircraft and, consequently, improving the modeling of performance.

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